



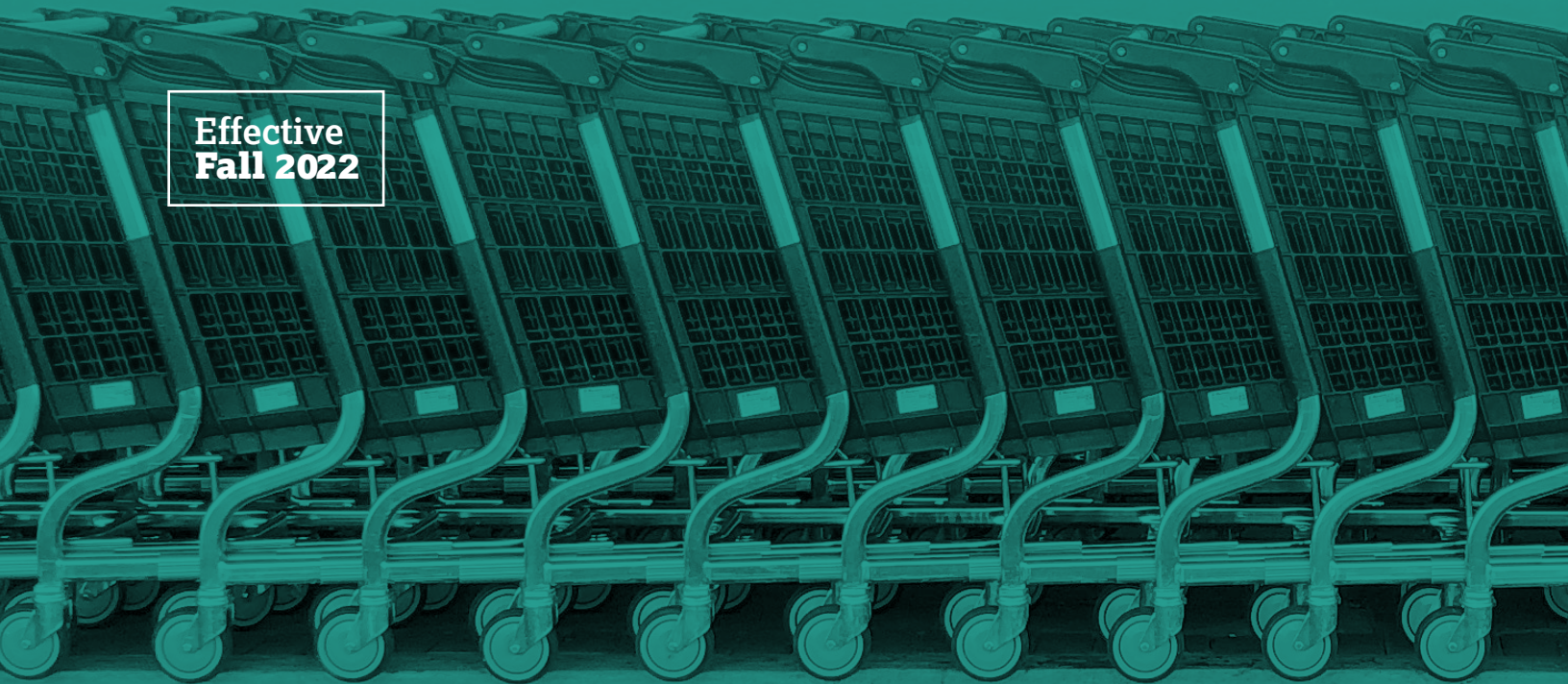
INCLUDES

- ✓ Course framework
- ✓ Instructional section
- ✓ Sample exam questions
- ✓ Classroom poster

AP[®] Microeconomics

COURSE AND EXAM DESCRIPTION

Effective
Fall 2022



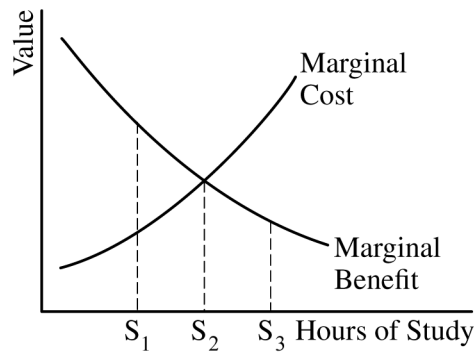
Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and the AP Microeconomics Exam and serve as examples of the types of questions that appear on the exam. After the sample questions are tables that show which skill, learning objective(s), and unit each question relates to. The answers to the multiple-choice questions are also provided.

Section I: Multiple-Choice

The following are examples of the kinds of multiple-choice questions found on the exam.

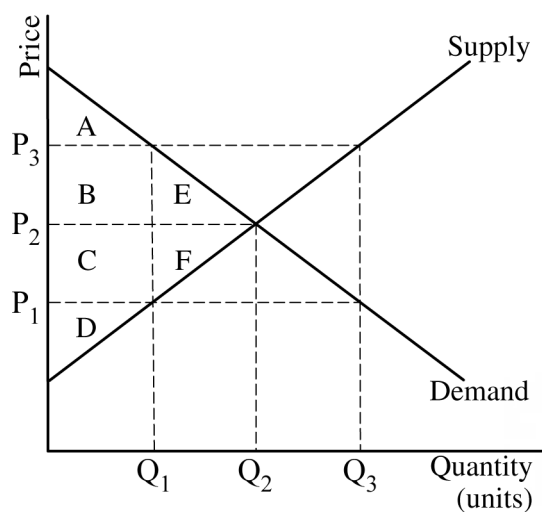
1. Individuals and societies must make choices regarding which goods and services will be produced and consumed because
 - (A) unlimited wants can only be satisfied by allocating resources efficiently
 - (B) resources are insufficient to satisfy the unlimited wants of people
 - (C) allocating scarce resources efficiently reduces the equitable distribution of goods
 - (D) income is not equally distributed to ensure that everyone can participate in the market
 - (E) a country cannot have a comparative advantage in the production of all goods



2. The graph above shows the marginal benefit that a student receives and the marginal cost the student incurs from additional hours of study. Which of the following is true?
 - (A) Beyond S_2 hours of study, there is no further benefit from devoting additional hours to study.
 - (B) At S_1 hours of study, the extra marginal cost of study time exceeds the extra marginal benefit of study time.
 - (C) The optimal number of hours that the student should study is S_2 hours.
 - (D) The rational individual should study S_3 hours.
 - (E) The marginal cost of studying decreases as more time is spent studying.

3. Which of the following occurs if the price of oranges is below the equilibrium price?
 - (A) There is a surplus of oranges.
 - (B) The quantity supplied of oranges is greater than the quantity demanded.
 - (C) The quantity supplied of oranges is greater than the quantity sold.
 - (D) The quantity purchased of oranges is greater than the quantity sold.
 - (E) The quantity demanded of oranges is greater than the quantity supplied.

Questions 4 and 5 refer to the graph below.



4. Total economic surplus will be maximized at which of the following price and quantity combinations?
- (A) P_3 and Q_3
 (B) P_3 and Q_1
 (C) P_2 and Q_2
 (D) P_1 and Q_1
 (E) P_1 and Q_3
5. Suppose the government sets a price floor at P_3 in the market. The resulting consumer surplus, producer surplus, and deadweight loss will be which of the following?

(A)

Consumer Surplus	Producer Surplus	Deadweight Loss
B + E	C + F	A + B

(B)

Consumer Surplus	Producer Surplus	Deadweight Loss
A + B	C + D	E + F

(C)

Consumer Surplus	Producer Surplus	Deadweight Loss
A	C + D + F	E

(D)

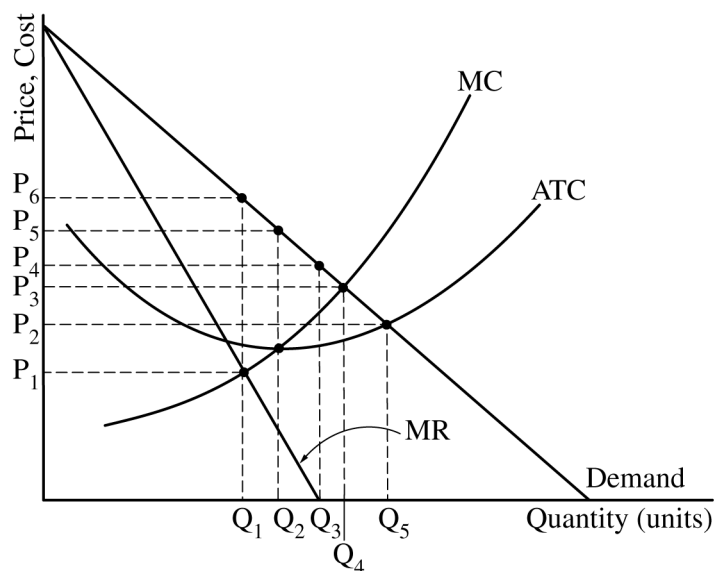
Consumer Surplus	Producer Surplus	Deadweight Loss
A + B + E	C + D + F	0

(E)

Consumer Surplus	Producer Surplus	Deadweight Loss
A	B + C + D	E + F

6. Assume that a 2 percent increase in the price of bologna causes a 5 percent decrease in the quantity demanded of cheese. What is the cross-price elasticity of demand between these goods, and how are these goods related?
- (A) Cross-price elasticity of demand equals -0.4 , and these goods are complements.
 - (B) Cross-price elasticity of demand equals $+0.4$, and these goods are substitutes.
 - (C) Cross-price elasticity of demand equals -2.5 , and these goods are complements.
 - (D) Cross-price elasticity of demand equals $+2.5$, and these goods are substitutes.
 - (E) Cross-price elasticity of demand equals -0.4 , and these goods are substitutes.
7. For a firm where labor is the only variable input, which of the following happens when diminishing returns set in?
- (A) Average variable cost begins to increase.
 - (B) Average product of labor begins to decline.
 - (C) Total product begins to decline.
 - (D) Marginal cost begins to increase.
 - (E) Average total cost begins to increase.
8. Which of the following indicates the presence of economies of scale as the quantity of output increases?
- (A) Short-run average variable cost decreases.
 - (B) Long-run average total cost decreases.
 - (C) Marginal cost decreases.
 - (D) Average fixed cost decreases.
 - (E) Marginal cost exceeds average total cost.

9. For a perfectly competitive firm, assume the price equals a rising marginal cost at 200 units of output. At this output, average total cost is \$8 and average variable cost is \$5. If the price is \$4, by how much can this firm reduce its losses by shutting down?
- (A) \$800
 - (B) \$600
 - (C) \$400
 - (D) \$200
 - (E) \$100
10. Which of the following is true for a perfectly competitive firm?
- (A) The firm is a price taker.
 - (B) The firm chooses its profit-maximizing quantity by equating price to average total cost.
 - (C) The firm faces a downward-sloping demand curve for its product.
 - (D) The firm's short-run supply curve is the same as its average variable cost curve above the minimum of the marginal cost curve.
 - (E) The firm can expect to earn positive economic profit in long-run equilibrium.
11. Which of the following is more likely to occur when there are high barriers to entry in an industry?
- (A) The firm(s) in the industry earn economic profits in the long run.
 - (B) The industry will be characterized by diseconomies of scale.
 - (C) The firm(s) in the industry are price takers.
 - (D) The firm(s) in the industry will charge a price equal to average total cost.
 - (E) The firm(s) will charge a price on the inelastic portion of the demand curve.



12. What are the profit- and revenue-maximizing quantities for the monopolist whose revenue and cost conditions are described by the graph above?

(A)	Profit-Maximizing	Revenue-Maximizing
	Q_5	Q_2
(B)	Profit-Maximizing	Revenue-Maximizing
	Q_4	Q_3
(C)	Profit-Maximizing	Revenue-Maximizing
	Q_3	Q_3
(D)	Profit-Maximizing	Revenue-Maximizing
	Q_1	Q_4
(E)	Profit-Maximizing	Revenue-Maximizing
	Q_1	Q_3

13. In long-run equilibrium, at which of the following output levels will a monopolistically competitive firm operate?
- (A) Where long-run average total cost is at a minimum
 - (B) Where price equals marginal revenue
 - (C) Where price equals average total cost
 - (D) Where marginal revenue equals average total cost
 - (E) Where marginal revenue equals average revenue
14. The equilibrium quantity of labor used in the production of a good will be the largest under which of the following conditions?
- (A) When the labor market is monopsonistic and the output market is perfectly competitive
 - (B) When the labor market is monopsonistic and the output market is a pure monopoly
 - (C) When the labor market is perfectly competitive and the output market is a pure monopoly
 - (D) When both the labor and output markets are perfectly competitive
 - (E) When laborers are allowed to collude in setting a price
15. Which of the following is true if the production of a good imposes external costs?
- (A) The market supply curve lies above the marginal social cost curve.
 - (B) An appropriate government policy would be to subsidize production of the good.
 - (C) A competitive market would not provide the good at all.
 - (D) A competitive market would produce more than the socially optimal amount.
 - (E) A persistent market surplus would occur.

Section II: Free-Response

The following are examples of the kinds of free-response questions found on the exam. Note that on the actual AP Exam, there will be one long free-response question worth 10 points and two short free-response questions, each worth 5 points.

1. On the island of Gratin, potatoes are produced in a perfectly competitive constant cost industry. The market for potatoes is currently in long-run equilibrium at the market price of \$5 per sack.
 - (a) Draw correctly labeled side-by-side graphs for the potato market and for farmer Lamo and show each of the following.
 - (i) The market equilibrium quantity and price, labeled Q_M and \$5, respectively
 - (ii) The quantity produced and the price for farmer Lamo, labeled Q_L and \$5, respectively
 - (b) Believing that the incomes of potato farmers are unfairly low, the government of Gratin imposes a binding price floor at \$7 per sack. The government will purchase any surpluses of potatoes. On your graph in part (a), show each of the following.
 - (i) The market quantity of potatoes that would be supplied at the price floor, labeled Q_F
 - (ii) The profit farmer Lamo earns, shaded completely
 - (c) If the absolute value of the price elasticity of demand is 0.92, would consumer spending on potatoes increase, decrease, or stay the same as a result of the price floor? Explain.
 - (d) How would the price floor affect the demand for labor by potato farmers?
 - (e) Potatoes, corn, and rice are all produced on the island of Gratin in perfectly competitive markets.
 - (i) Assume corn is a substitute in production (an alternate output) for potatoes. How would setting the price floor on potatoes affect the equilibrium price and quantity of corn?
 - (ii) As a result of an increase in the price of rice, the demand for potatoes increased. Are rice and potatoes substitutes or complements in consumption?

2. Two discount stores, Discount Delight and Bargain Floor, sell a popular brand of athletic shoes. They are considering including these shoes in their upcoming sale. The relevant payoff matrix appears below, with the first entry in each cell indicating Discount Delight's profit from shoe sales and the second entry in each cell indicating Bargain Floor's profit. The two firms know all the information in the payoff matrix and do not cooperate.

		Bargain Floor	
		Includes	Does Not Include
Discount Delight	Includes	\$150, \$180	\$200, \$100
	Does Not Include	\$100, \$250	\$240, \$220

- If Discount Delight does not include the shoes in its sale but Bargain Floor includes them, what will Bargain Floor's profit be?
- Does either player have a dominant strategy? If so, identify the player and the strategy.
- Using numbers from the table, explain why both stores choosing not to include the shoes in their respective sales does **not** correspond to a Nash equilibrium.
- Identify the Nash equilibrium strategy for each store.
- If the two stores could cooperate in choosing their strategies, would the outcome change? Explain.

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Unit
1	B	1.A	MKT-1.A	1
2	C	2.A	CBA-2.B	1
3	E	2.A	MKT-4.B	2
4	C	2.A	MKT-4.A	2
5	E	3.A	POL-1.A	2
6	C	2.C	MKT-3.E	2
7	D	2.A	PRD-1.A	3
8	B	1.A	PRD-1.A	3
9	D	3.C	PRD-2.A, CBA-2.C	3
10	A	1.A	PRD-3.A	3
11	A	1.A	PRD-3.B	4
12	E	2.A	PRD-3.B	4
13	C	2.A	PRD-3.B	4
14	D	2.A	PRD-3.A, PRD-4.C	5
15	D	2.A	POL-3.A	6

Free-Response Question	Question Type	Learning Objectives	Unit
1	Long	MKT-3.B, MKT-3.D, MKT-3.E, POL-4.A, PRD-3.A, PRD-4.B	2, 3, 5, 6
2	Short	PRD-3.C	4

The scoring information for the questions within this course and exam description, along with further exam resources, can be found on the [AP Microeconomics Exam Page](#) on AP Central.



Scoring Guidelines

Question 1: Long

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 - (ii) The quantity produced and the price for farmer Lamo, labeled Q_L and \$5, respectively
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Scoring Guidelines for Question 1: Long

10 points

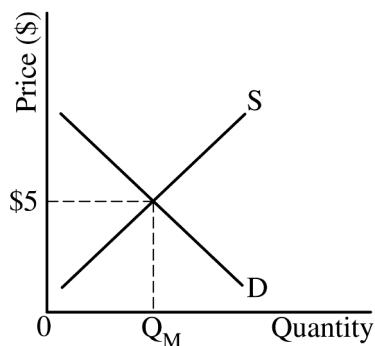
Learning Objectives: MKT-3.B MKT-3.D MKT-3.E POL-4.A PRD-3.A PRD-4.B

- (a) Draw a correctly labeled graph of the potato market that shows the market equilibrium quantity and price labeled Q_M and \$5, respectively.

1 point

4.A

Task type: Create graphs or visual representations

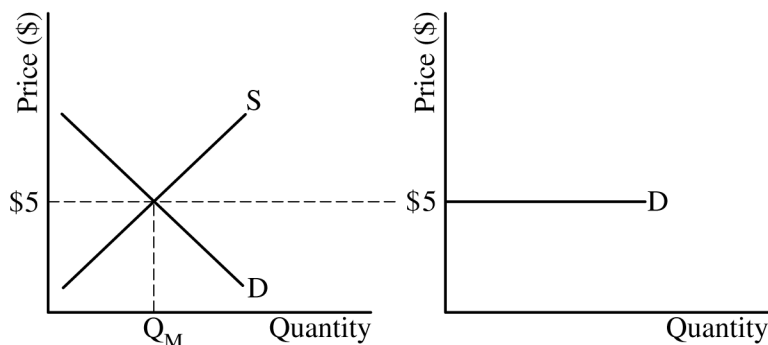


Draw a correctly labeled graph for farmer Lamo next to the graph of the potato market that shows a horizontal demand curve for Lamo coming from the market equilibrium price of \$5.

1 point

4.A

Task type: Create graphs or visual representations

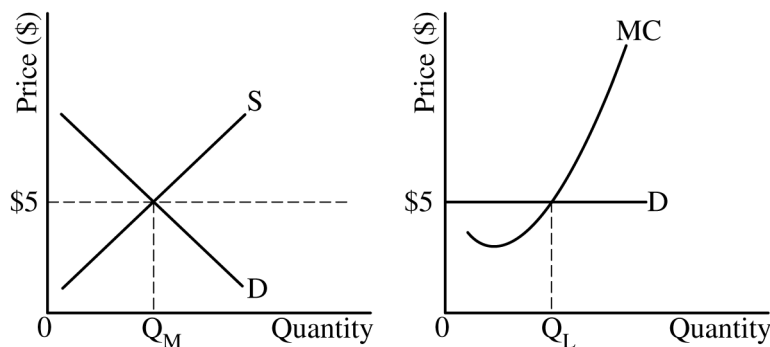


Show the profit-maximizing quantity Q_L , where $P = MC$.

1 point

4.B

Task type: Create graphs or visual representations

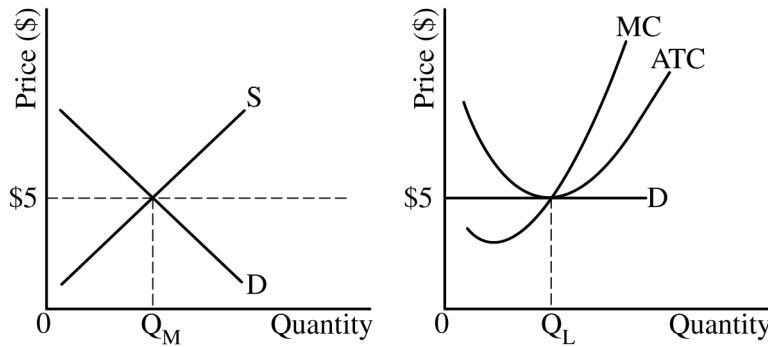


Show the ATC curve tangent to Lamo's demand curve at Q_L .

1 point

Task type: Create graphs or visual representations

4.B



Total for part (a)

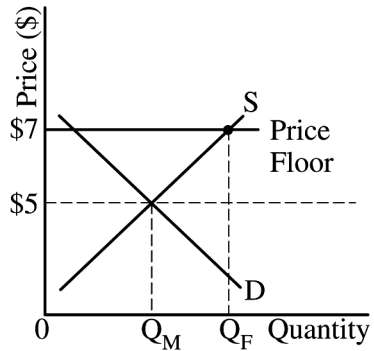
4 points

- (b) (i) On the graph from part (a), show the market quantity of potatoes that would be supplied at the price floor, labeled Q_F .

1 point

4.C

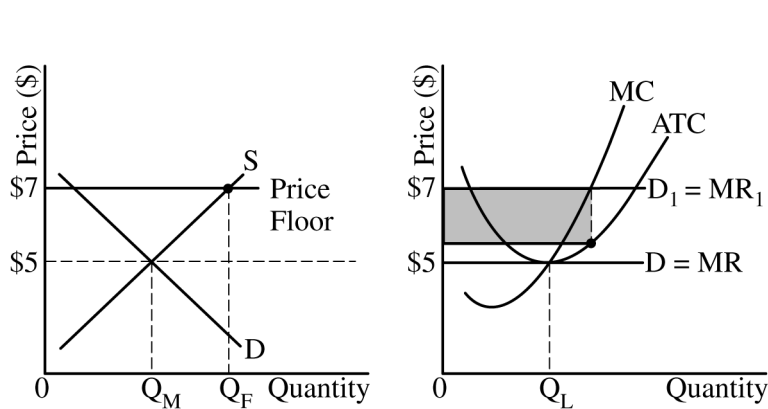
Task type: Create graphs or visual representations



- (ii) On the graph from part (a), show the profit farmer Lamo earns with the price floor, shaded completely.

1 point

4.C



Total for part (b)

2 points

- (c) State that consumer spending on potatoes will increase and explain that because the demand for potatoes is inelastic, the price increase will result in an increase in total spending. (It is also acceptable to explain that because demand is inelastic, the percentage increase in price outweighs the percentage decrease in the quantity demanded of potatoes, resulting in an increase in total spending on potatoes.)

1 point

3.C

Task type: Explain

(d)	State that the demand for labor will increase and/or the demand curve will shift to the right. <i>Task type: Make an assertion</i>	1 point 3.B
(e)	(i) State that the equilibrium price of corn will increase and the equilibrium quantity of corn will decrease. <i>Task type: Make an assertion</i>	1 point 3.B
	(ii) State that potatoes and rice are substitutes in consumption. <i>Task type: Make an assertion</i>	1 point 2.A
		Total for part (e) 2 points
		Total for question 1 10 points

Question 2: Short

2. Two discount stores, Discount Delight and Bargain Floor, sell a popular brand of athletic shoes. They are considering including these shoes in their upcoming sale. The relevant payoff matrix appears below, with the first entry in each cell indicating Discount Delight's profit from shoe sales and the second entry in each cell indicating Bargain Floor's profit. The two firms know all the information in the payoff matrix and do not cooperate.

		Bargain Floor	
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Discount Delight	Includes	\$150, \$180	\$200, \$100
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- If Discount Delight does not include the shoes in its sale but Bargain Floor includes them, what will Bargain Floor's profit be?
- Does either player have a dominant strategy? If so, identify the player and the strategy.
- Using numbers from the table, explain why both stores choosing not to include the shoes in their respective sales does **not** correspond to a Nash equilibrium.
- Identify the Nash equilibrium strategy for each store.
- If the two stores could cooperate in choosing their strategies, would the outcome change? Explain.

Scoring Guidelines for Question 2: Short

5 points

Learning Objectives: PRD-3.C

- | | | |
|------------|--|------------------------------|
| (a) | State that Bargain Floor's profit will be \$250.
<i>Task type: Perform numerical analysis</i> | 1 point
1.C |
| (b) | State yes and identify that including the shoes in its sale is the dominant strategy for Bargain Floor.
<i>Task type: Perform numerical analysis</i> | 1 point
1.C |
| (c) | Explain that it is not a Nash equilibrium because Bargain Floor has an incentive to move to including the shoes in its sale, earning \$30 more (\$250 > \$220).
<i>Task type: Explain</i> | 1 point
2.C |
| (d) | Identify the Nash equilibrium strategy for each store as choosing to include the shoes in the sale.
<i>Task type: Perform numerical analysis</i> | 1 point
2.C |
| (e) | State yes and explain that by both agreeing to not include the shoes in their sales, they would both increase their profits.
<i>Task type: Explain</i> | 1 point
3.C |

Total for question 2 5 points